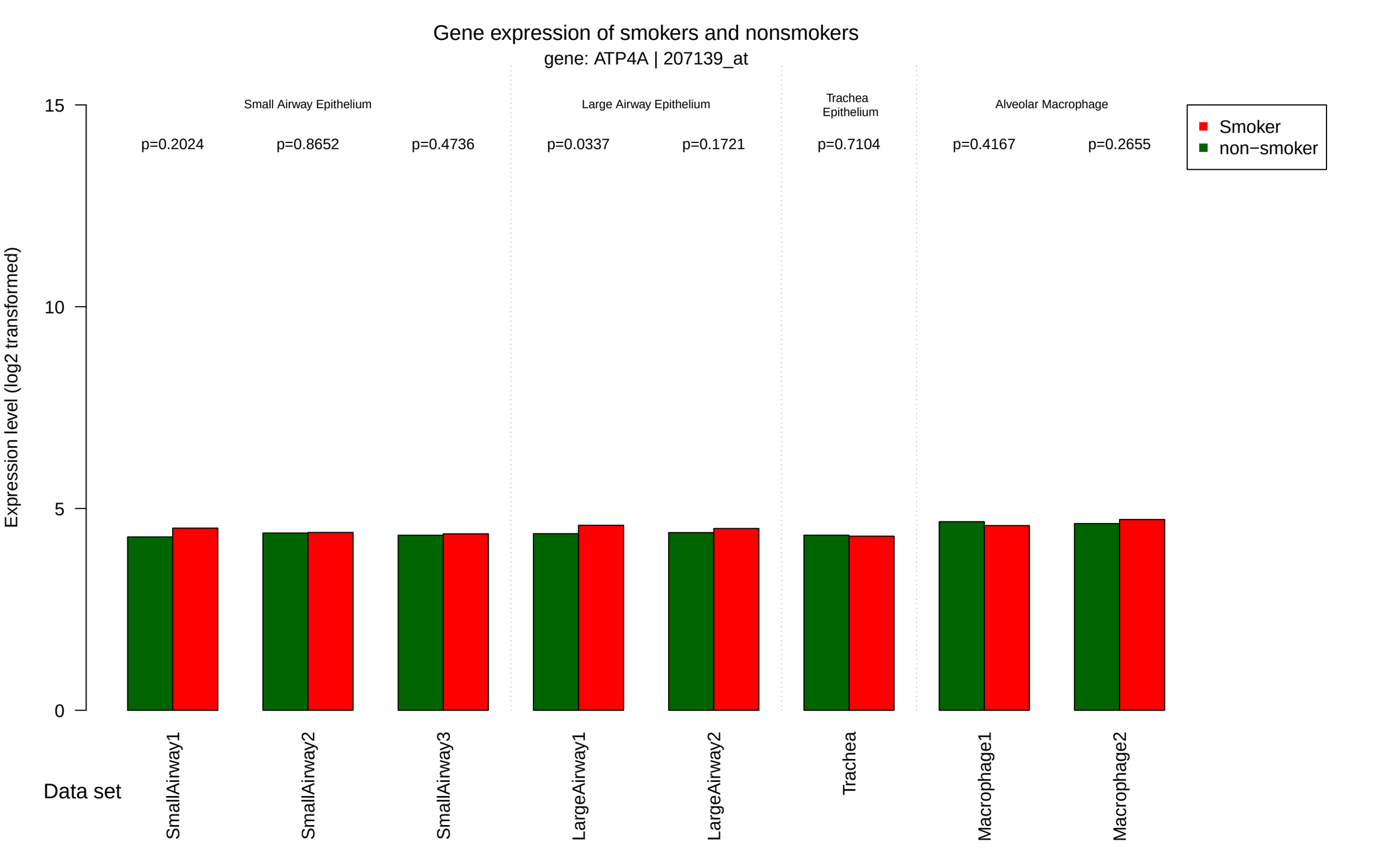
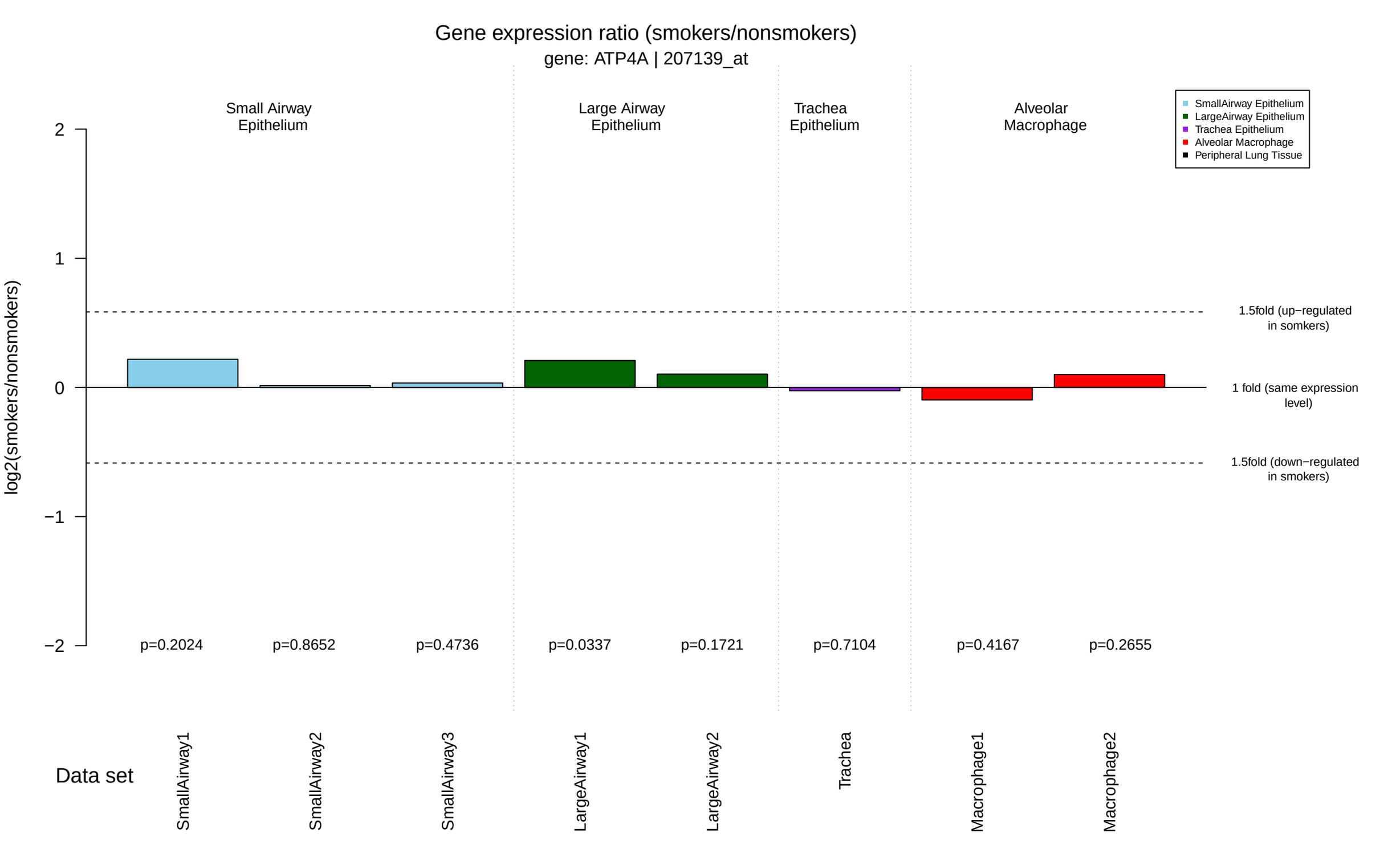
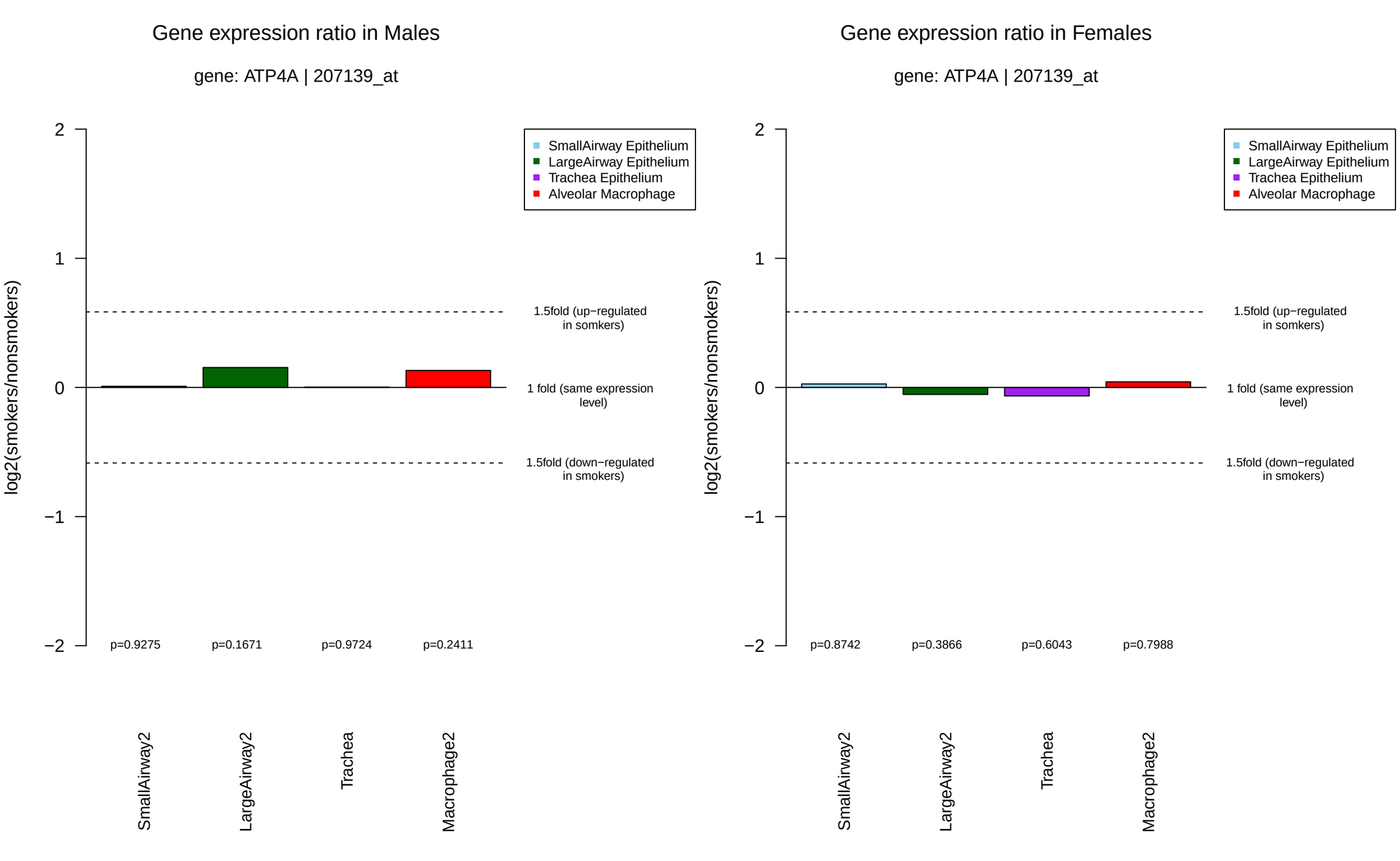
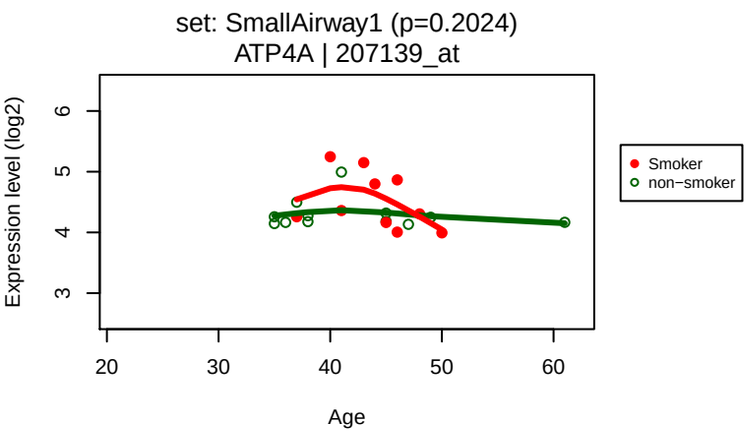
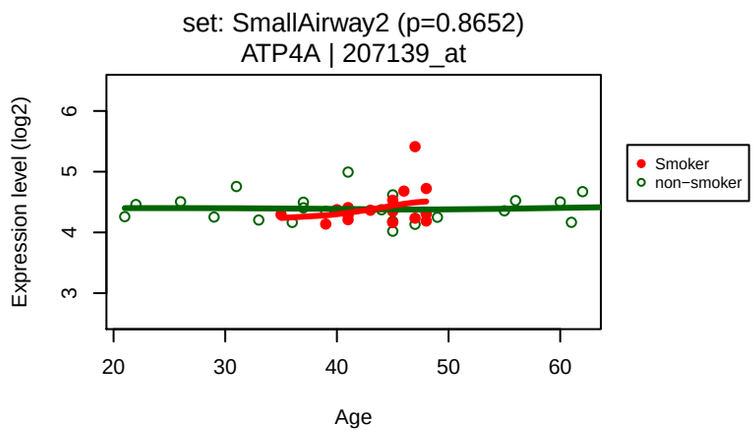
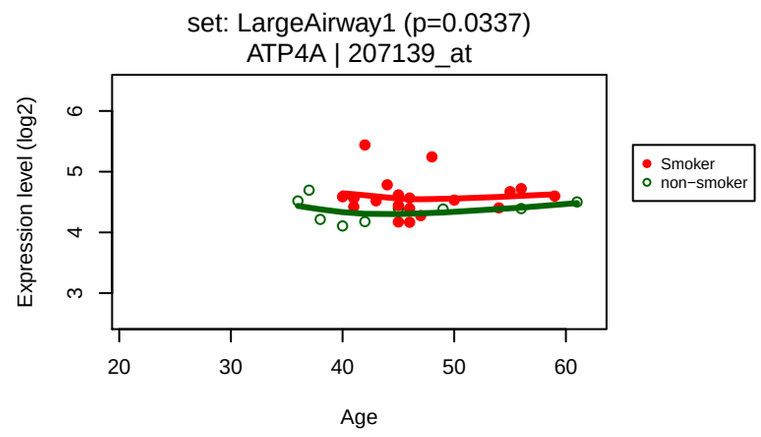

Gene expression of smokers and nonsmokers
gene: ATP4A | 207139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2024
p=0.8652
p=0.4736
p=0.0337
p=0.1721
p=0.7104
p=0.4167
p=0.2655
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

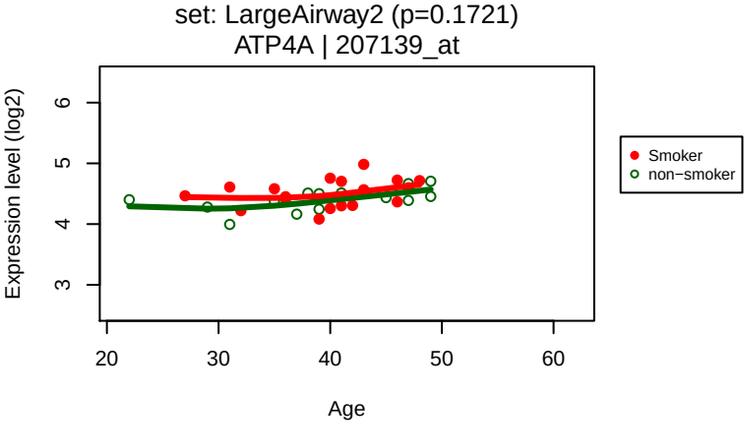

Gene expression ratio (smokers/nonsmokers)
gene: ATP4A | 207139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2024
p=0.8652
p=0.4736
p=0.0337
p=0.1721
p=0.7104
p=0.4167
p=0.2655
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

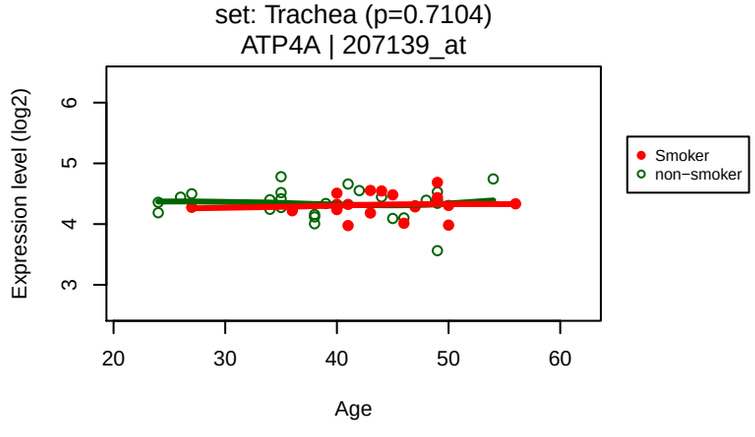

Gene expression ratio in Males
gene: ATP4A | 207139_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9275
p=0.1671
p=0.9724
p=0.2411
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP4A | 207139_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8742
p=0.3866
p=0.6043
p=0.7988
SmallAirway2
LargeAirway2
Trachea
Macrophage2

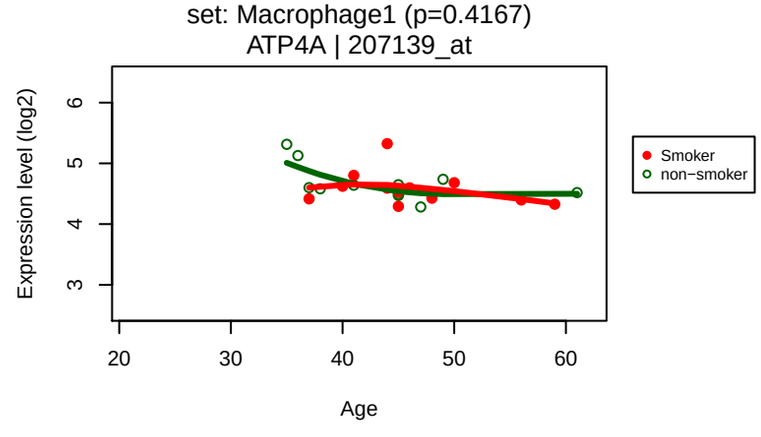

set: SmallAirway1 (p=0.2024)
ATP4A | 207139_at
Expression level (log2)
Age
Smoker
non−smoker

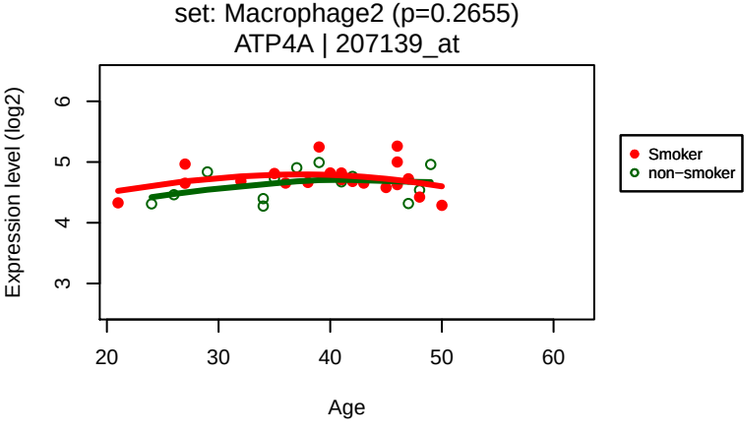

set: SmallAirway2 (p=0.8652)
ATP4A | 207139_at
Expression level (log2)
Age
Smoker
non−smoker

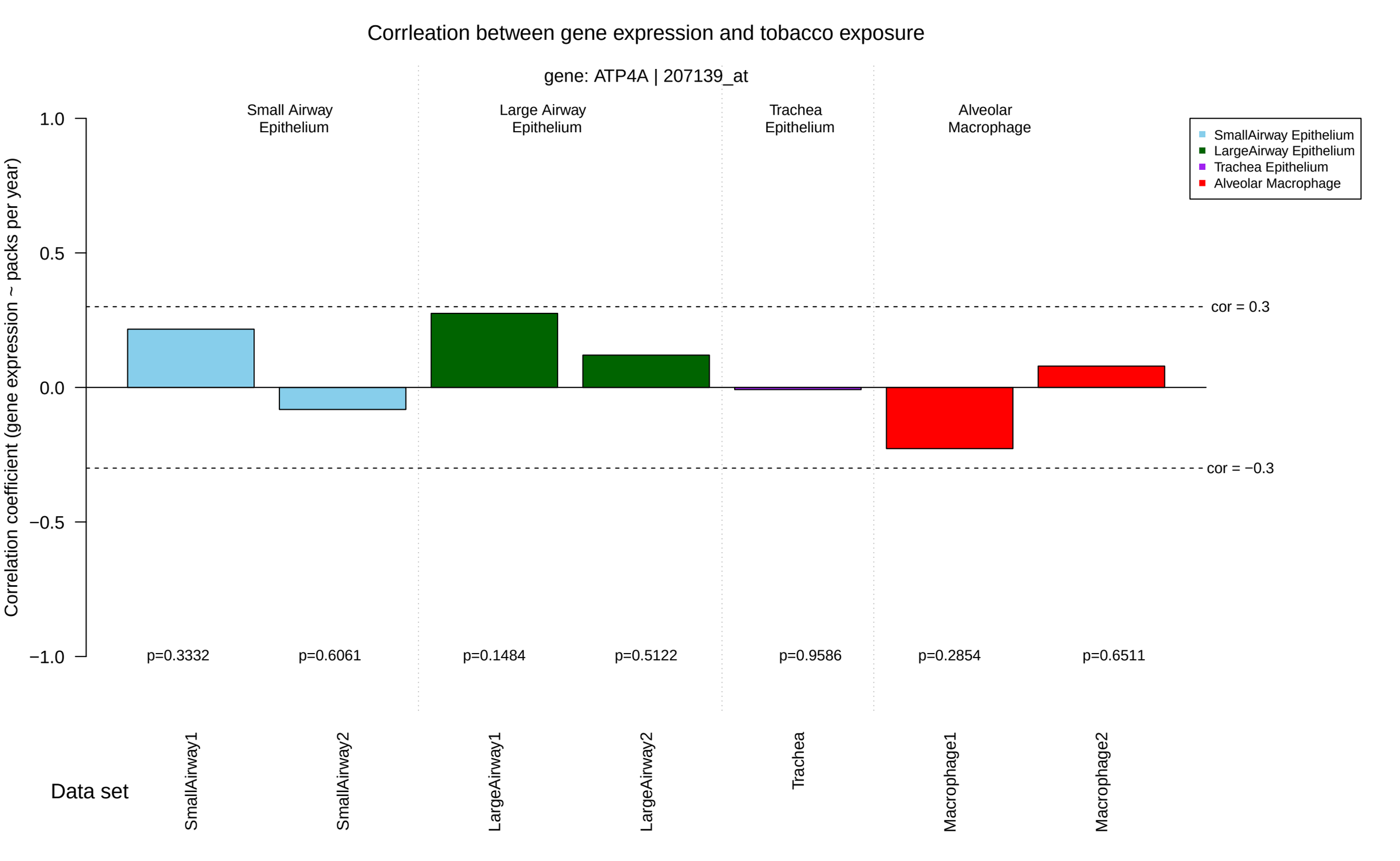

set: LargeAirway1 (p=0.0337)
ATP4A | 207139_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1721)
ATP4A | 207139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7104)
ATP4A | 207139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4167)
ATP4A | 207139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2655)
ATP4A | 207139_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP4A | 207139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3332
p=0.6061
p=0.1484
p=0.5122
p=0.9586
p=0.2854
p=0.6511
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2